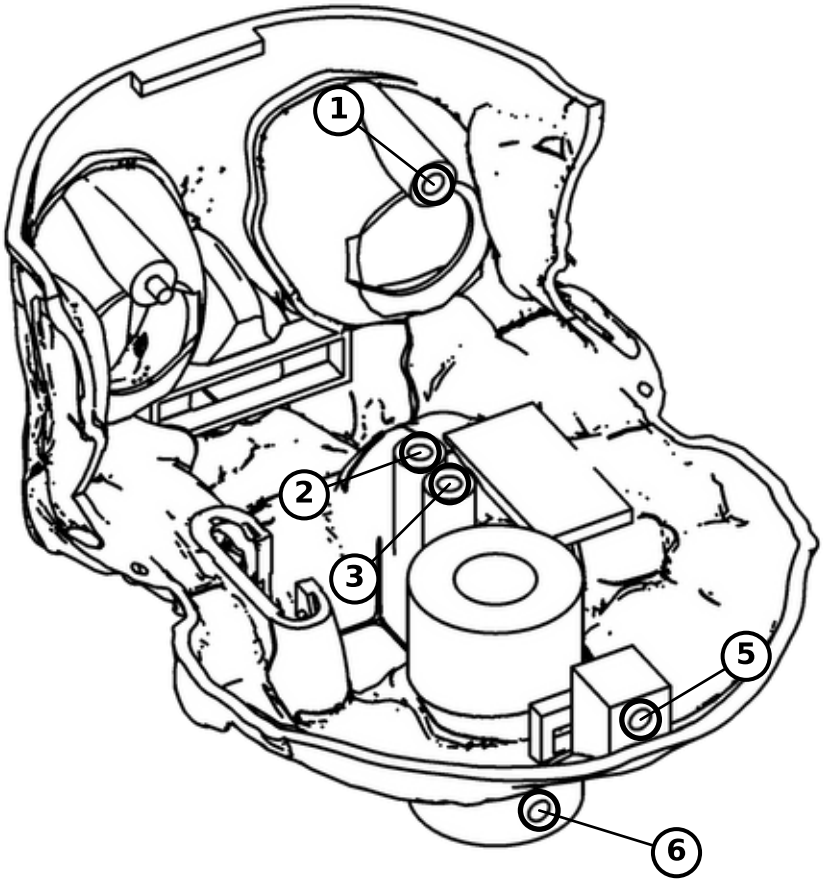
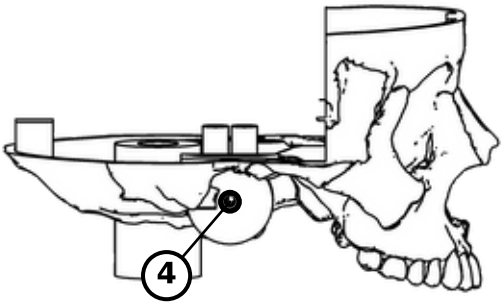
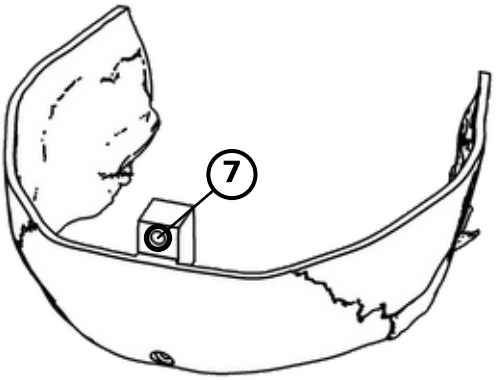
1a Inserts M3 ×7



face, side

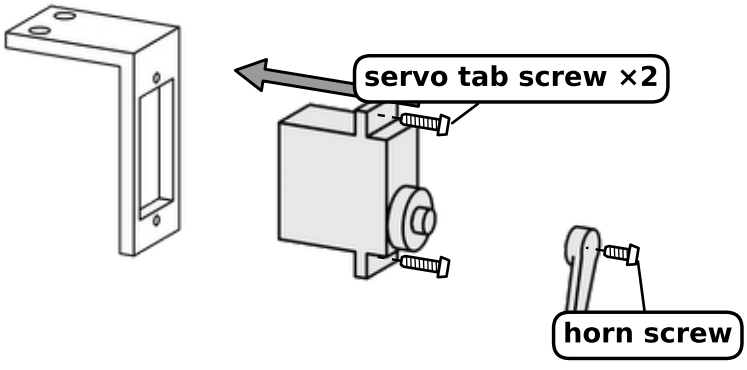


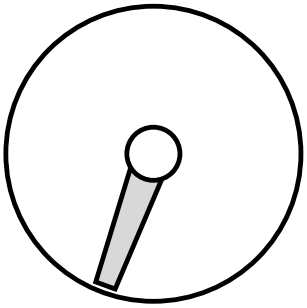
back cover



Heat-set flush, square to the hole.

1b Servo + bracket



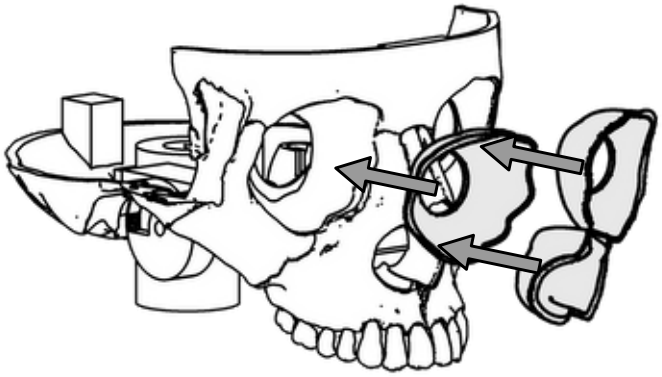


horn down

Tab screws in. Centre the servo (90°), then the horn, pointing down the jaw arm.

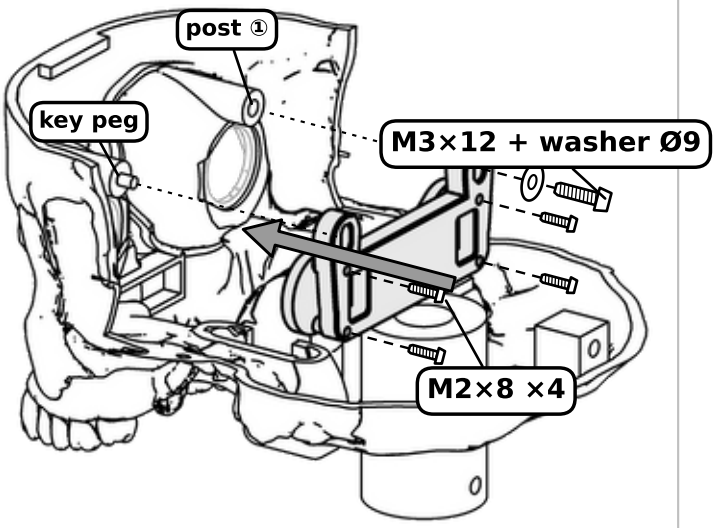
M2×8 ×4	M2×16 ×2	M3×8 ×3	M3×12 ×3	M3×20 ×1	Ø9 washer ×1	M3 washer ×1	insert ×7

2 Liners



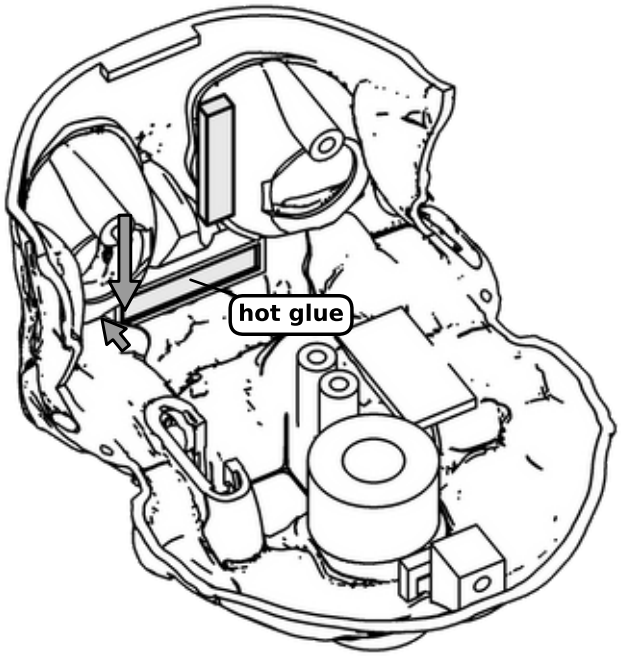
2–3 dabs of glue on the seat.
Push in from the front, lip flush.

3 Eyes



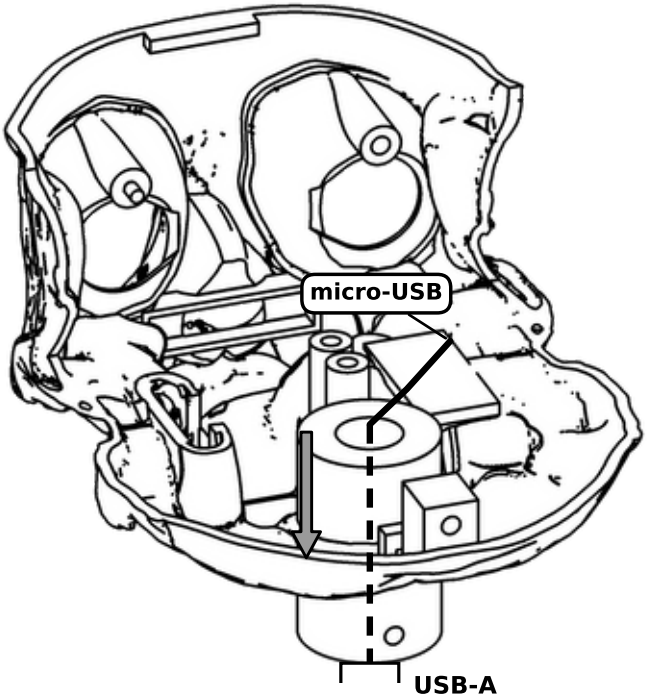
Plug the LCD cables in first. Bolt hole on the post ①, other hole on the key peg. Centre the screens.

4 Radar



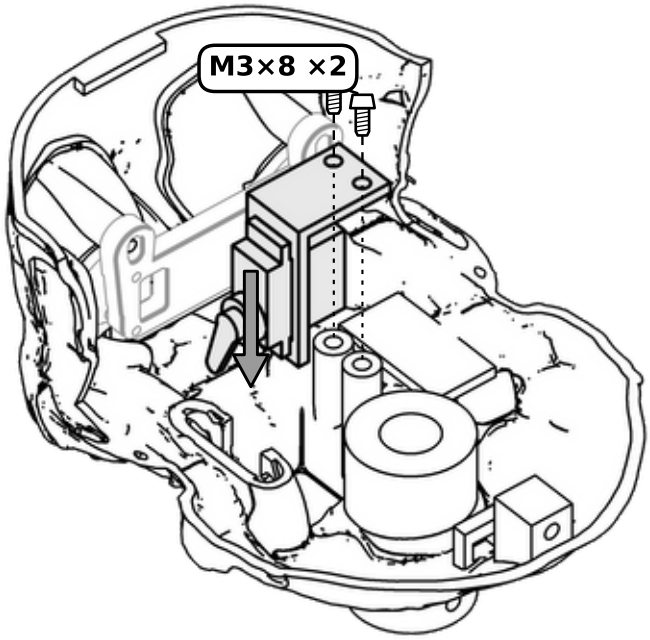
On edge down the slot between the eyes, then flat and forward into its frame. Hot glue.

5 USB cable



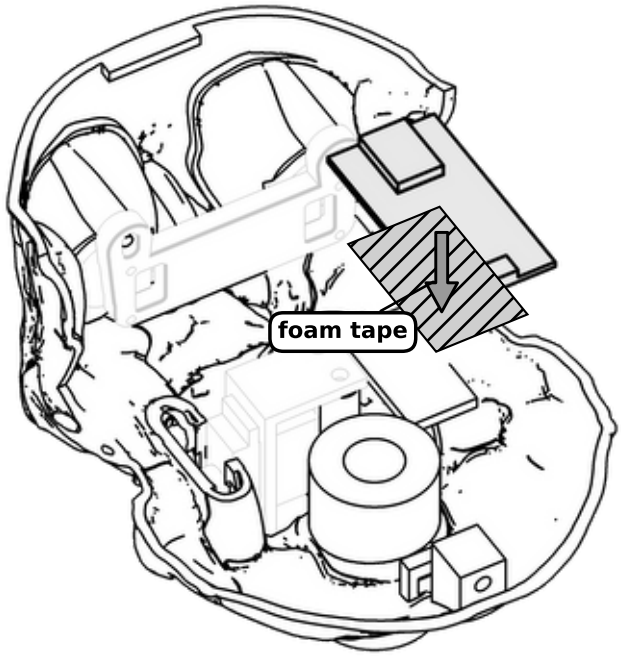
USB-A end down the neck bore from inside. Micro-USB end stays inside.

6 Servo



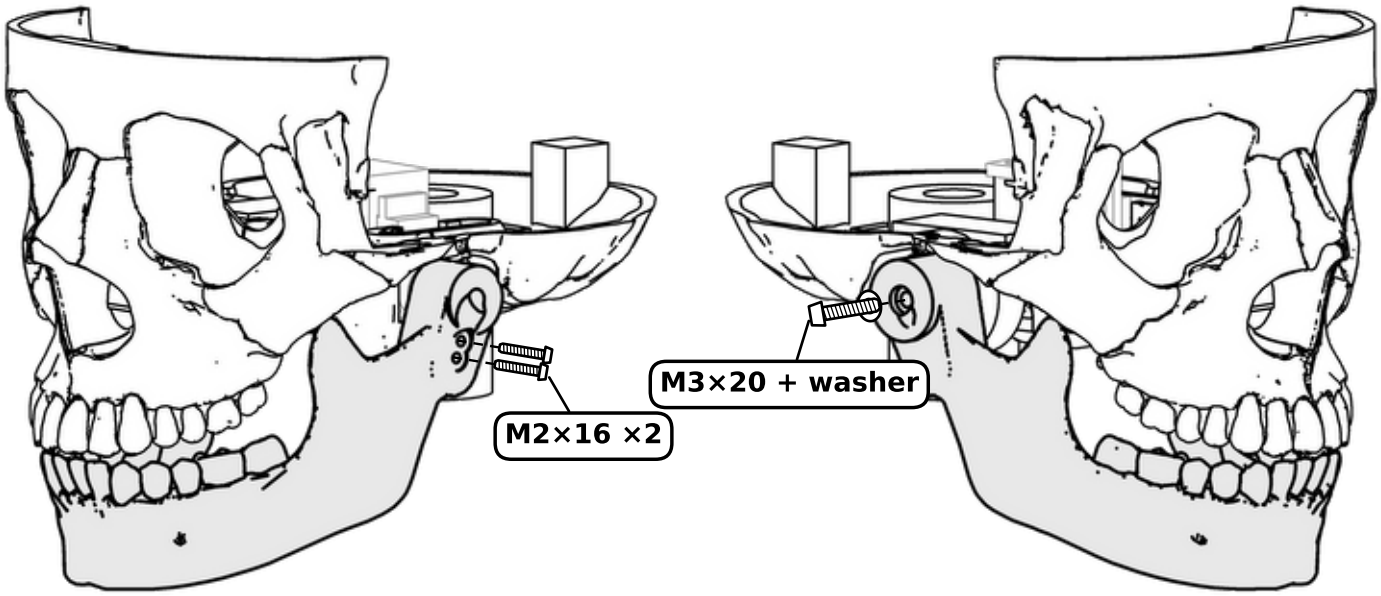
Gear tower down the slot in the cup wall.
Bolts into inserts ② ③.

7 ESP32



No tray: foam tape, flat behind the eye bar (solder side covered). Plug in: LCD ×2, radar, servo, USB.

8 Jaw



Left: M2×16 from outside the jaw into the horn arm.
Right: pivot bolt + washer into insert ④. Snug, free to swing.

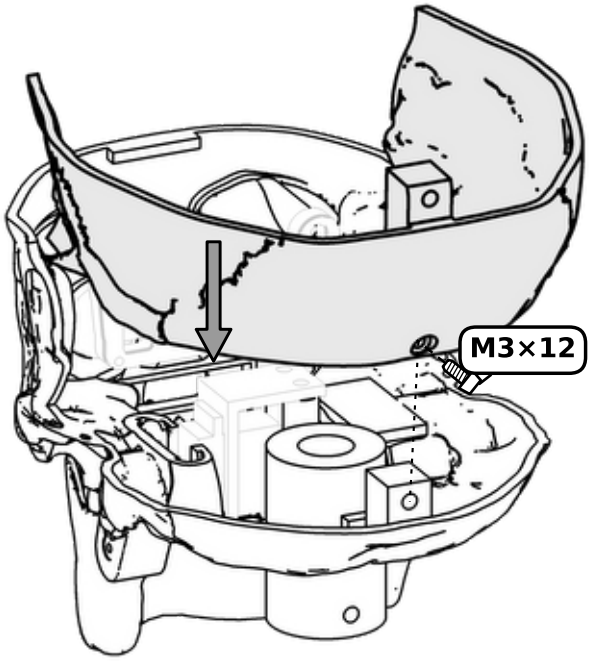
9 Bench test



- [ ] eyes
- [ ] jaw
- [ ] radar
- [ ] sound

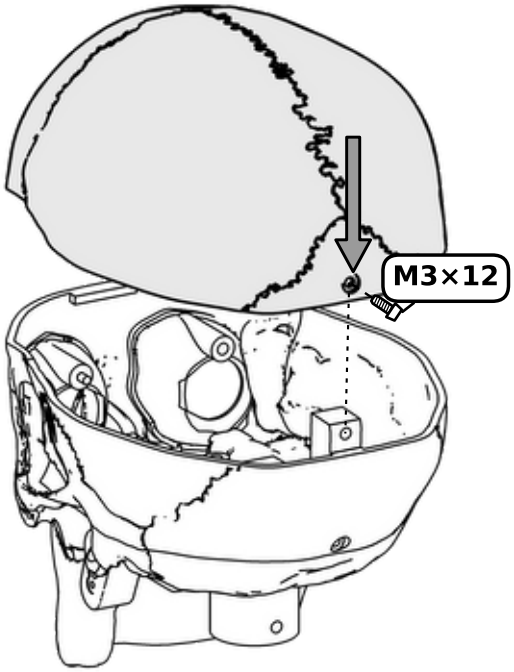
Power up before closing the skull.

10 Back cover



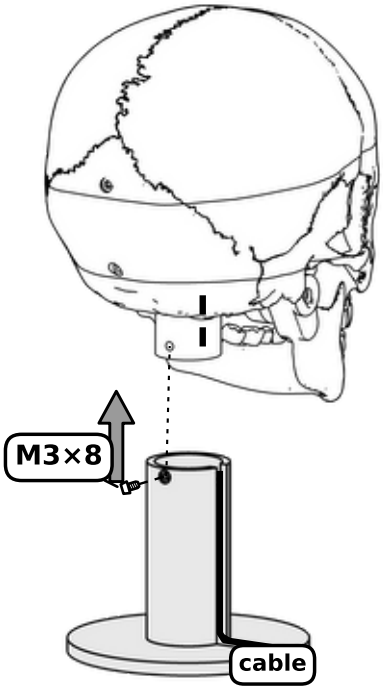
Drop it onto the floor lip.
Screw low at the back, into insert ⑤.

11 Cap



Straight down onto the cut.
Screw high at the back, into insert ⑦.

12 Stand



Cable in the column slot. Push the stand onto the neck, screw from behind into insert ⑥.